



TM415

Introduction to mapp Axis

Requirements

Training modules	TM210 – Working with Automation Studio TM400 – Introduction to motion control
Software	Automation Studio 4.9 Automation Runtime 4.91 and later mapp Motion Technology Package
Hardware	X20 controller ACOPOS devices / X20 stepper motor module

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1 Introduction

The B&R drive solution is **fully integrated into Automation Studio**. This enables consistent configuration and commissioning of all drive components.

This training module describes the steps in Automation Studio to develop and implement a motion control solution.

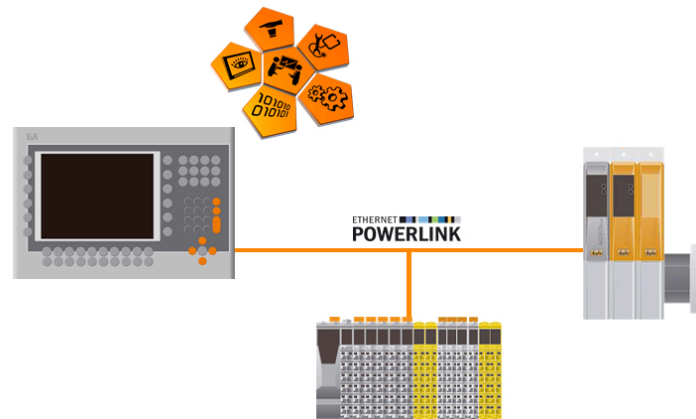


Figure 1: The integrated automation solution from B&R

During this training course, participants will take a closer look at the role played by the various elements of a motion control solution. The diagnostic tools included in Automation Studio provide an ideal environment for efficient testing and commissioning. The exercises in this training module are designed to help participants become familiar with the fundamentals of working with integrated motion control. Automation Help provides support during configuration and detailed descriptions of all elements.

1.1 Learning objectives

- Participants will get an overview of B&R motion products.
- Participants will gain basic knowledge about wiring and troubleshooting drive hardware, displayed on the hardware.
- Participants will know where to find required information in Automation Help.
- Participants will be able to add and configure an axis in Automation Studio.
- Participants will be able to operate and diagnose an axis.
- Participants will know the necessary parameters required to operate a motor with a B&R drive.
- Participants will know the concept of regulation and understand how autotuning is carried out. Participants will also be familiar with controller parameters that can be readjusted manually.
- Participants will be able to execute various options for simulations for drives in Automation Studio.
- Participants will categorize important information for a support request at B&R.
- Participants will know possibilities for displaying axis information without Automation Studio.

1.2 Symbols and safety notices

Unless otherwise specified, the symbol descriptions and safety notices listed in "TM210 - Working with Automation Studio" apply.

1.3 B&R online courses

The B&R online courses provide lessons for a wide range of topics. Because the courses are interactive, they allow content to be learned effectively.

To help you find the B&R online courses you need more quickly, the various courses on the website are assigned to different training categories and are based on the modular training concept.

[B&R online courses](https://www.br-automation.com/en/academy/online-courses/) <https://www.br-automation.com/en/academy/online-courses/>)



2 The hardware

At the beginning, it is important to deal with the hardware that is used. Which connections does the device have? How should it be wired?

Hardware information sources

Product-related information are available in the **user's manual** and in **Automation Help**.

The user's manual is available on the B&R website in the product area. A device can be quickly found by entering its serial number (see nameplate) in the search field on the website.

In Automation Help, product-related information is stored in chapter "Hardware".

Further information:

- www.br-automation.com

Wiring

There is an overview for each B&R hardware that shows all connections and labels. The pinout of the individual connectors are clearly displayed in a separate section.

The hardware includes labels and stickers with clear instructions for wiring the pins. This makes troubleshooting easier.

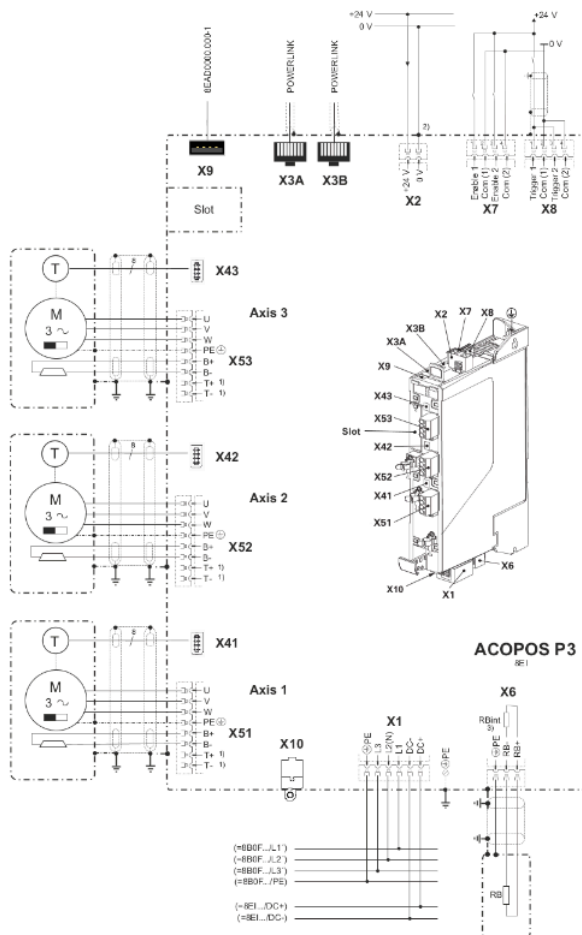


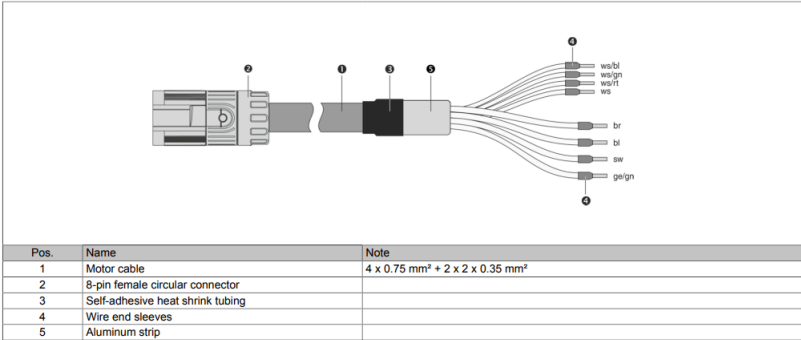
Figure 2: The ACOPOS P3 user's manual, technical data, pinout

If a motor is connected to an ACOPOS servo device, the order of the cable ends must be observed. Motor phases U, V and W must be correctly connected to the ACOPOS terminal.

Failure to adhere to the motor cable pinout may result in incorrect direction of rotation or irreparable damage to motor components.

An improperly connected motor may perform uncontrolled movements!
This behavior cannot be compensated by the software!
The wiring must be carried out properly according to the official B&R description!

Cable construction



Pinout

Circular connector	Pin	Name	Function
	A	U	Motor connection U
	B	V	Motor connection V
	C	W	Motor connection W
	+	PE	Protective ground conductor
	1	T+	Temperature +
	2	T-	Temperature -
	3	B+	Brake +
	4	B-	Brake -
	5	---	---

Cable diagram

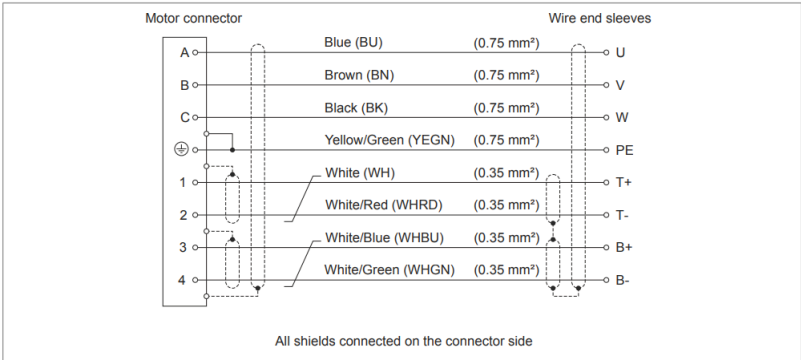


Figure 3: The ACOPOS P3 user's manual, technical data, motor cable 0.75 mm²

Hardware \ Motion control \ <Device> \ Technical data \ <Device type> \ Wiring
Hardware \ Motion control \ <Device> \ Technical data \ Accessories \ Cables or motor cables

Exercise: Obtaining information about the hardware

The goal of this exercise is to help participants to use Automation Help and the hardware section of the user's manual. Participants should find the cable diagram and the color coding of the motor cable and the pinout of the device used. In addition, the hardware used should be checked for wiring errors.

- 1) Open the B&R website and search for the device used (e.g. ACOPOS P3 user's manual, serial number for related information)
- 2) Open the hardware section of the device being used in Automation Help.
- 3) Find the cable diagram and color coding of the B&R motor cable.
- 4) Find the pinout of the device used.

3 The software

This chapter describes the communication between controller and drive and the configuration of the various drive parameters. It also explains which interface the motion control application and diagnostic tools use to communicate with a drive axis.

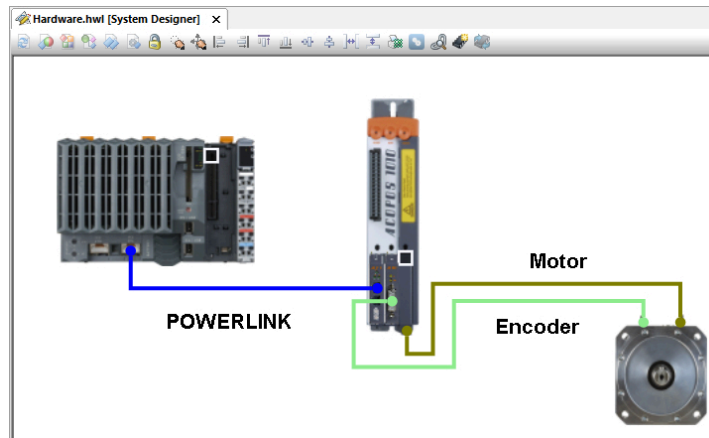


Figure 4: A motion control system as seen in System Designer. Data exchange between the controller and ACOPOS servo drive via POWERLINK. The ACOPOS servo drive is connected to the motor via a motor cable and an encoder cable.

The System Designer image shows the components required for a drive configuration.

The X20 controller is connected to the ACOPOS servo drive via POWERLINK. The ACOPOS servo drive is connected to the motor using a motor cable and an encoder cable.

The motor cable transfers the power to the motor. The encoder cable is used for position feedback to the ACOPOS servo drive.

Configuration in Automation Studio

Procedure	Description
Configuration	The hardware for the motion control system is configured in Automation Studio. This involves, for example, adding an ACOPOS servo drive in the Physical View or in System Designer.
Configuration	The drive configuration must be adapted to match the actual components being used.
Diagnostic and testing tools	System Diagnostics Manager and mapp Cockpit offer various diagnostic and commissioning aids. They can be used to start positioning sequences, issue other commands and record data for detailed analysis, see 6 "Commissioning and diagnostics" on page 25 .
Simulation	Automation Studio allows you to simulate motors and servo drives, see 8 "Simulation options" on page 41 .
Software development	Drive solutions from B&R are implemented with mapp Motion. For more information, see training modules "TM416 – Motion control: Basic functions" and "TM417 – Motion control: Multi-axis functions".

Table 1: Overview of configuration options in Automation Studio

3.1 Drive parameters and drive communication

Drive parameters

Drive parameters can generally be divided into two general groups.

The first group includes all the parameters required for the **hardware configuration**. These are device-specific parameters required for real motion and real hardware operation.

The second group includes all the parameters required for the **positioning sequences**. These are generally adjusted to the movement and can be homed and processed on any device. The combined structure is referred to as the axis or axis group reference.

Parameters for the hardware configuration	Parameters for positioning sequences
Motor characteristic values	System of units
Encoder interfaces	Motion parameters
Supply voltage	Speed/acceleration values
Controller values	Axis limit values

Table 2: Example listing of how drive parameters can be grouped

Drive communication

The controller communicates with the ACOPOS servo drive via POWERLINK. The data transmitted by the controller controls the drive.

Module McAcDrv, a software object called the ACOPOS driver, provides libraries (e.g. McAxis) with function blocks. McAcDrv is used to operate ACOPOS drive hardware and to provide services for cyclic and acyclic communication with the ACOPOS hardware.

The ACOPOS servo drive provides the motor with setpoint values that are returned via an encoder interface.

The data needed for drive communication can be divided into two general categories.

Drive parameters and commands	Status data
The drive parameters are transferred from the controller to the devices in the ACOPOS devices. Positioning commands are transferred to the devices in the ACOPOS devices.	Devices in the ACOPOS devices use status data to report the state of the drive control and positioning procedures back to the controller.

Table 3: Differentiating between parameters, commands and status data

The operating system and the parameters needed to configure the drive are transferred to the drive automatically. This also happens after the connection to a drive is lost or when a drive is replaced by a new device. No other steps are required to update the drive.

mapp function blocks (MpAxis library) and PLCOpen function blocks (McAxis library) serve as programming interfaces in the drive application.

3.2 Software concept and software interfaces

Regardless of which particular drive components are used, axes are all addressed in the same way in the software. A **component** is used to manage each axis (later linked to a servo drive, frequency inverter, stepper motor drive or hydraulic drive). Several configuration objects in addition to the component are available for expansions. Since it is independent of the motion control technology used, the same motion control application can always be used to control an axis.

The corresponding component is used by both the motion control application and the diagnostic environment to address the corresponding drive axis. Real and virtual axes are available for single-axis applications, couplings via electronic gearbox and cams.

TM1111 describes path-controlled movements, i.e. when individual axes are interdependent or affect each other (e.g. robots with linear kinematics, CNC machines).

We will take a closer look at the single-axis concept in the following pages.

Real axis

This component is used to operate a real servo drive with a motor and position encoder, for example. The servo drive generates setpoints for the motor, and motor movement is monitored using the position encoder.

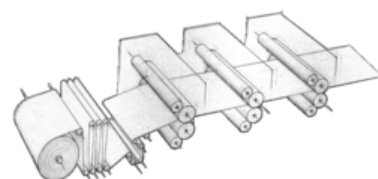


Figure 5: Motor driving a winder drive

Virtual axis

In addition to real axes, ACOPOS also offers the option of operating a virtual drive. This drive works solely as a type of "setpoint generator" that generates values for position and speed. It is operated in the same way as a real axis. A real axis can be coupled to the position of a virtual axis.

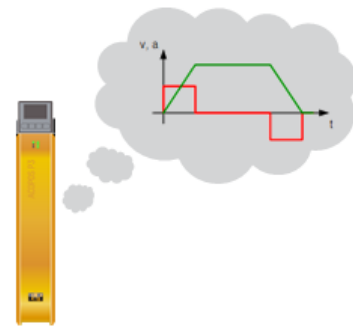


Figure 6: Virtual axis e.g. in an ACOPOS P3 device

Purely virtual axis

Purely virtual axes are axes without physical characteristics. The position, speed and acceleration values are calculated on the controller using a setpoint generator (McProfGen).

A purely virtual axis can be integrated in an Automation Studio project in the Configuration View using configuration file PureVAX.

Programming the purely virtual axis can be done using PLCopen function blocks from the McAxis library.

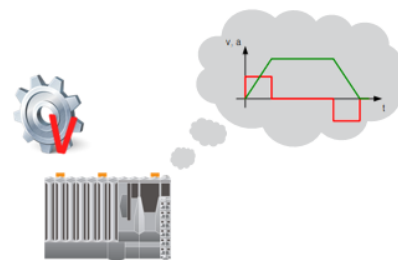


Figure 7: Purely virtual axis on the PLC

A purely virtual axis must be switched on like a real axis. It does not have to be homed to start a movement (the position value is initially set to 0 when the controller starts up).

Purely virtual axes can, for example, be used as:

- Coupling master for one or more hardware axes
- Simulation on the controller without drive hardware
- Centralized setpoint generation on the controller for various hardware



Motion control \ mapp Motion \ Concept \
Motion control \ mapp Motion \ Programming \ Application program \ Libraries \ Core \ McPureVAX

4 My first project

This chapter will help participants create their projects step by step – from adding a drive to Automation Studio to performing the first movements in mapp Cockpit. At the end, they will complete an exercise.

The following steps must be carried out:

- 1) Create an Automation Studio project with an X20 controller.
- 2) "Getting started" axis, see [4.1 "Configuring the single axis" on page 11](#)
 - 1) [4.1.1 "Adding configuration file" on page 11](#)
 - 2) [4.1.2 "Adjusting the axis object" on page 12](#)
 - 3) [4.1.3 "Adding ACOPOS servo drive" on page 12](#)
 - 4) [4.1.4 "Performing a hardware-specific configuration" on page 15](#)
 - 5) [4.1.5 "Performing an offline installation using CompactFlash" on page 16](#)
- 3) "Getting started" mapp Cockpit



Diagnostics and Service \ mapp Cockpit \ Getting started with mapp Cockpit \

- Preparing to use mapp Cockpit
- Accessing the web-based mapp Cockpit HMI application
- Interacting with mapp components in the web-based HMI application

4.1 Configuring the single axis

The following steps can also be performed together with "Getting started" in Automation Help or with the "Getting started" tutorial.



Motion control \ mapp Motion \ Getting started \ Axis

4.1.1 Adding configuration file

The required configuration files for an axis can be added in the motion section of the Configuration View in Automation Studio.

By setting the filters in the Toolbox, the required files can be located quickly and efficiently.

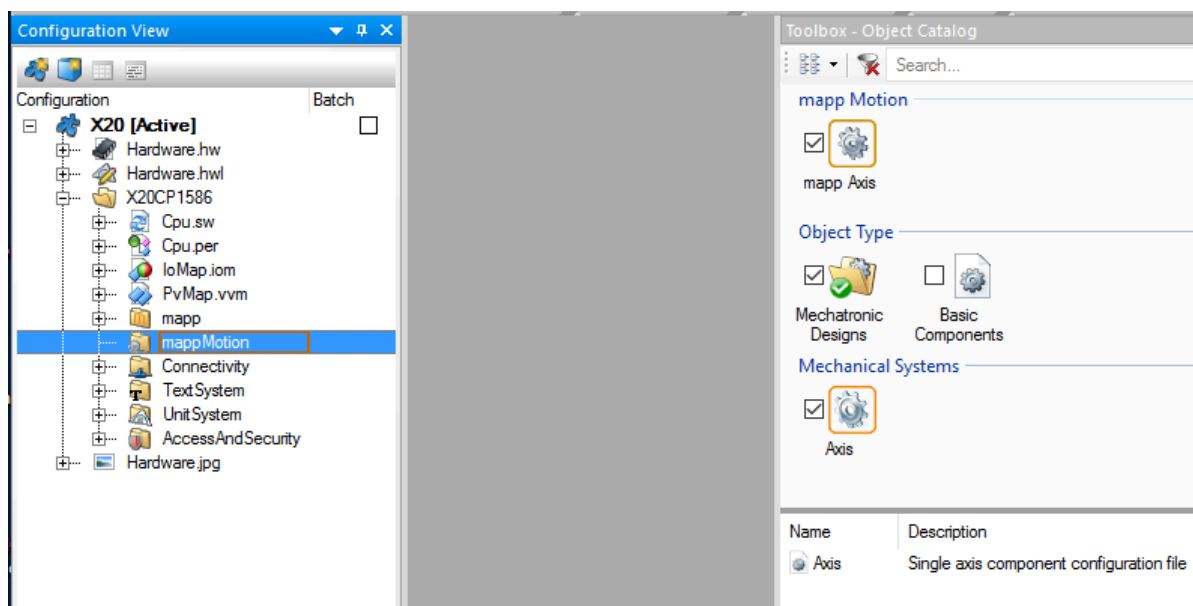


Figure 8: Adding an axis from the Toolbox by double-clicking (mapp Motion section must be selected beforehand)

4.1.2 Adjusting the axis object

When an axis object is added in the Configuration View, it is given a default name. Name "gAxis_1" can be changed as needed.

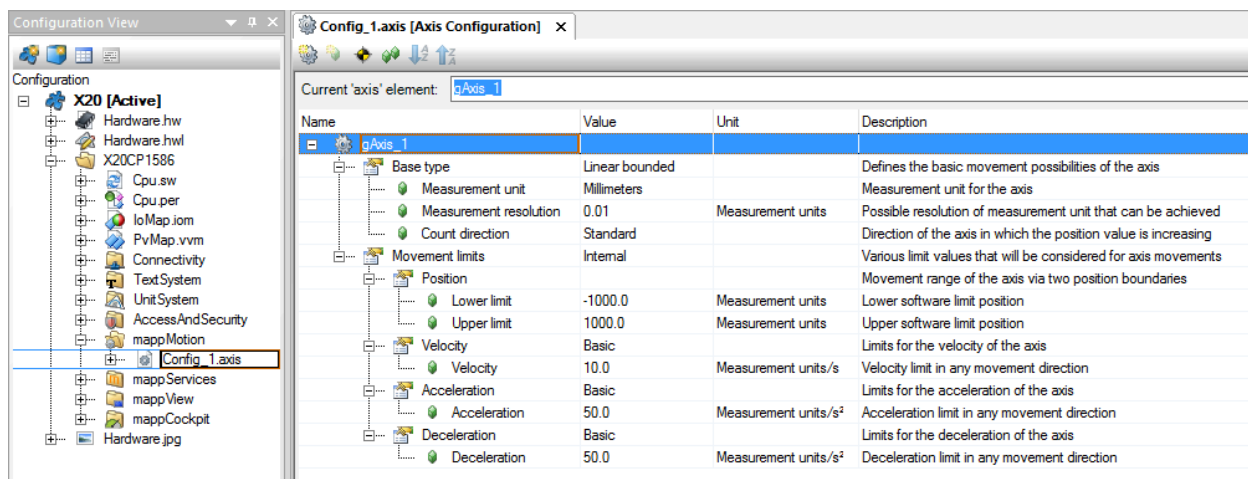


Figure 9: Double-clicking on Config_1.axis (in the Configuration View) brings you to the configuration for an axis

Multiple axis objects with different names can be created here. A single axis is needed for this "Getting started" section.

Further information

- [5.1 "Configuration modules" on page 18](#)

4.1.3 Adding ACOPOS servo drive

After creating an Automation Studio project with an X20 controller, an ACOPOS servo drive is connected via POWER-LINK.¹

¹ Other devices from the B&R motion control portfolio are also added to the Physical View via hardware access.



Codian or Comau robots with ACOPOS drive hardware can be added via the specific robot wizard starting with Automation Studio 4.10 and mapp Motion 5.16. The required configuration objects and the drive hardware are automatically added to the project and the robot is fully parameterized. This step can be skipped! Continue with Structured Text motion commands to create a program.

The hardware can be selected in the Hardware Catalog and then dragged and dropped to the appropriate interface in the Physical View (or in System Designer).

ACOPOSmicro is used as an example here.

However, thanks to the system's extensive scalability ("Scalability+"), this "Getting started" tutorial can also be applied to ACOPOS drive, ACOPOSmulti, ACOPOS P3, ACOPOSmotor, etc.

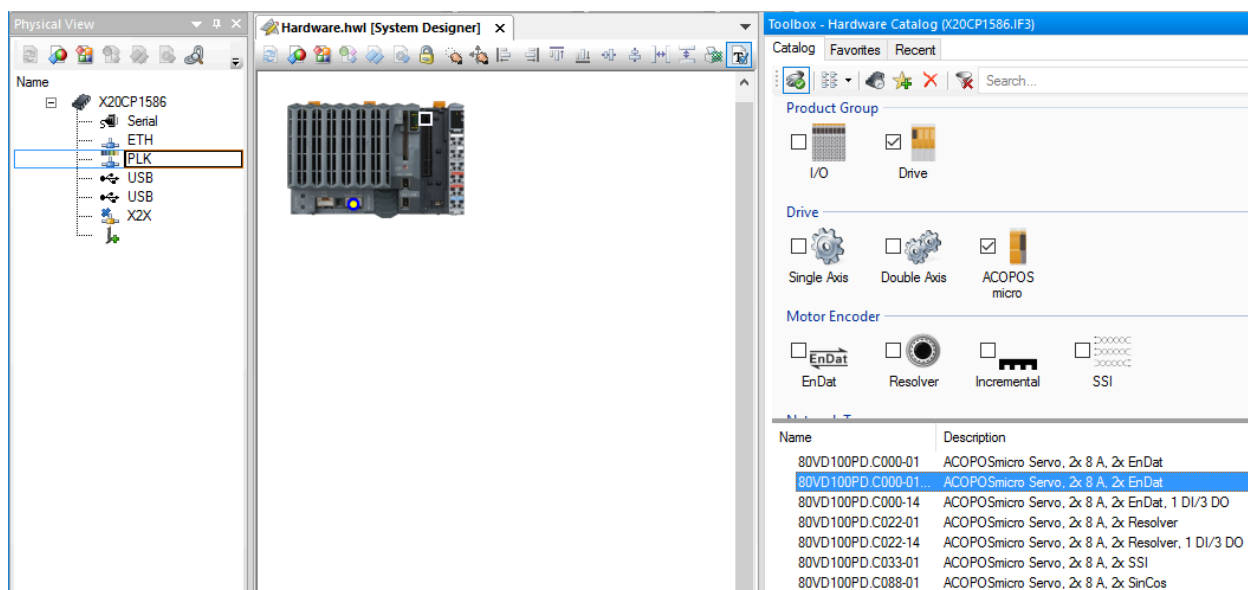


Figure 10: View before hardware is added from the Toolbox

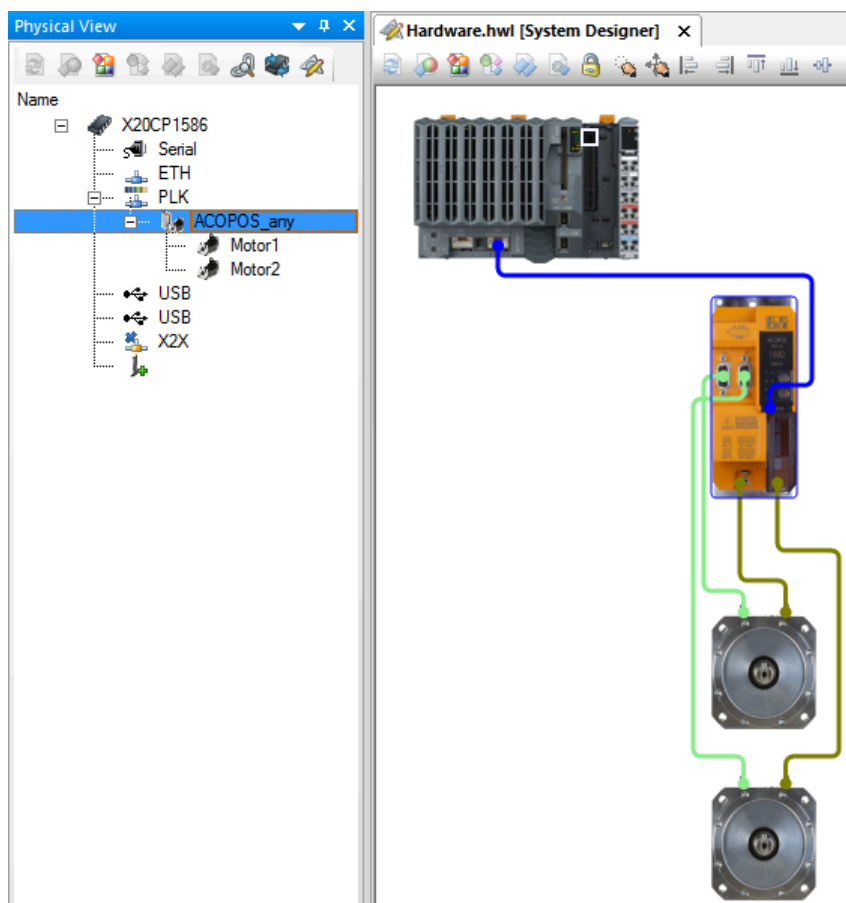


Figure 11: Attaching hardware to the POWERLINK interface by double-clicking

After working through the Getting started documentation, the drive is visible in the Physical View and in System Designer. It is possible to add or replace encoder interfaces by selecting the slots² on the drive.

After adding the drive in the Physical View, Automation Studio automatically created all the necessary libraries in the Logical View. For a description on this, see 5 "Components of the motion control system" on page 18.

Embedded parameter chip

The EnDat encoder system includes nonvolatile, maintenance-free data memory to store all of the data required to operate the drive motor. B&R stores motor data, among other things, so that the correct motor parameters and limit values are always automatically available to the ACOPOS drive system. This preprogrammed data is automatically transferred to the servo drive when the system is started.

In addition to assistance during commissioning, routine service work is also simplified, and motors can be replaced without having to take extra time to set parameters.



Motion control \ mapp Motion \ Concept \ Drive technology \ Components of a drive system \ Position encoder \ EnDat absolute encoder

Setting the POWERLINK node number



The node number on the ACOPOS must match the node number in Automation Studio.

² Additional encoder options and plug-in cards can be connected to available slots on ACOPOS and ACOPOSmulti systems.

The devices connected via POWERLINK receive node numbers from Automation Studio in ascending order. These node numbers can be changed in the shortcut menu in the Physical View or in System Designer.

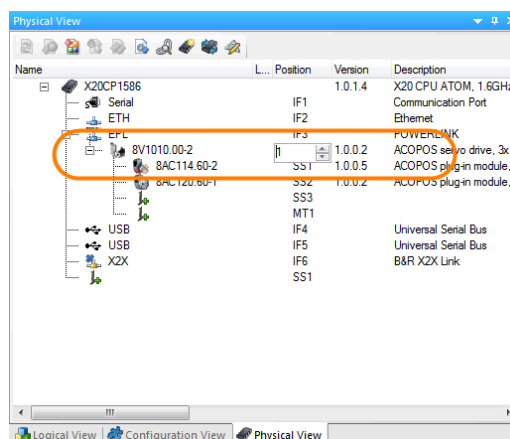


Figure 12: Changing the node number in the Physical View

4.1.4 Performing a hardware-specific configuration

This is where the previously created axis object is assigned to the hardware and the first necessary configuration settings are made. These settings can be adjusted as needed (e.g. homing method, simulation, limit values, etc.).

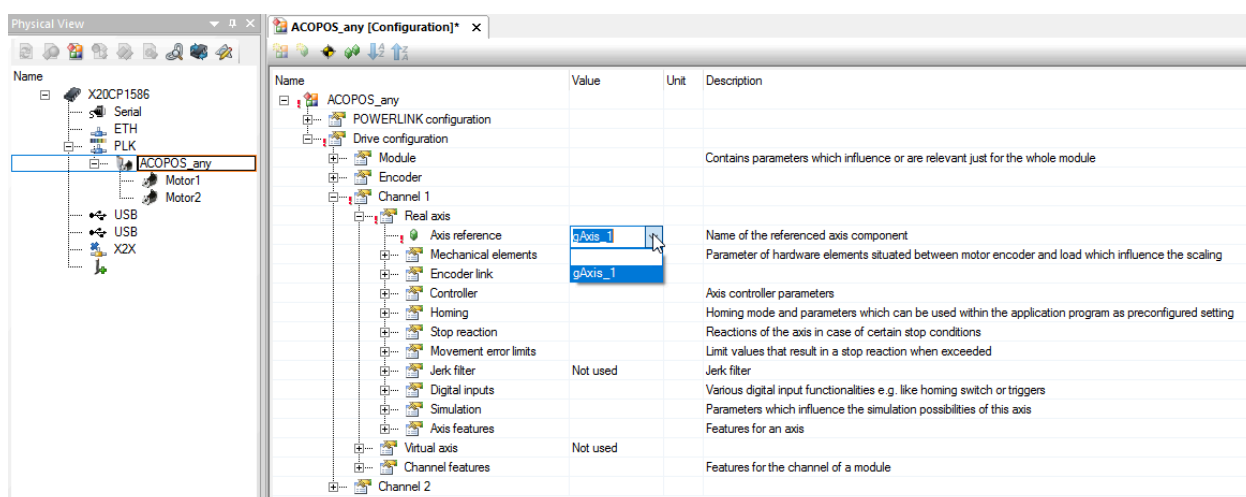


Figure 13: Double-clicking the hardware in the Physical View brings you to the hardware-specific configuration. The axis is assigned here.

If axes are connected to actual hardware, you must check the wiring ("Digital inputs"). If a system such as ETAlight 410 is used, the quick stop function should be disabled on all channels as the reference switch for homing would otherwise unintentionally trigger this emergency stop function.

The project is prepared and can be generated. After restarting the CPU, you can check in the Logger whether the system was started correctly.

The program can be tested using the Watch window and [6 "Commissioning and diagnostics"](#) on page 25.

Further information:

- [5.1 "Configuration modules"](#) on page 18

4.1.5 Performing an offline installation using CompactFlash

The modified hardware configuration can be loaded to the target system using a CompactFlash card (offline installation) or via the online connection. When the X20 controller is restarted, all parameters will then be transferred to the drive. The "R/E" LED lights up solid green on the drive once the transfer and a restart have taken place.

Further information:

- In the respective user's manual
- Under the "Hardware" section in Automation Help

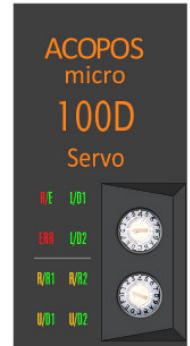


Figure 14: ACOPOS micro servo drive LED status indicators



Project management \ Hardware management \ Physical View \ Editing operations \ Changing the node number
Getting started \ Creating programs in Automation Studio \ Example project for a target system with CompactFlash

4.2 Preparing a drive and performing movements

The hardware for the X20 controller and drive was configured in Automation Studio and transferred to the controller. Now the first motor movement can be started.

Open mapp Cockpit

mapp Cockpit can be used to put any drive axis or axis groups in Automation Studio.



Diagnostics and service \ mapp Cockpit \ Getting started with mapp Cockpit

- Preparing the project for use with mapp Cockpit
- Accessing the web-based mapp Cockpit HMI application

Switch on controller AutoTune and perform homing procedure

After opening mapp Cockpit, several commands are available. The corresponding command is selected and executed. The controller parameters for initial use can be determined using the autotuning procedure. If this step is omitted, a lag error can occur when starting a movement at high loads or instability can occur at low loads and with small motors. The sequence should be observed, see ["Determine the controller settings using autotuning"](#) on page 35.

- Speed controller (possibly with filter for encoder feedback: "Filter time" or resonances: "Notch")
- Position controller
- Testing the controller
- Feed-forward control (optional)

Command "**Power on**" is used to enable the position control and the motor holds the current position.

Command "**Home**" is used to create a reference to the zero point and the system of units. This then makes it possible to move to a position or move a certain distance.

Switching on and homing only have to take place once; it is then possible to perform any number of movements.

Perform a movement

After preparation (Tune, Power, Home) has been completed, the drive is ready to perform movements. For example, commands "Move Absolute", "-Additive" and "-Velocity" can be used to rotate the motor clockwise and "Abort" can be used to stop a movement.

A detailed description of all options in mapp Cockpit is covered later in section, see [6 "Commissioning and diagnostics"](#) on page 25.



Motion control \ mapp Motion \ Diagnostics

Exercise: First movement with "Getting started"

Objectives:

- Project creation
- Configure the ACOPOS servo drive.
- Opening mapp Cockpit
- Move the axis.

The individual steps are listed at the beginning of this chapter. The exact procedure is described in the subchapters. In addition, 2 "Getting started" chapters can be consulted.



Motion control \ mapp Motion \ Getting started \ Axis
Diagnostics and Service \ mapp Cockpit \ Getting started with mapp Cockpit

- Preparing to use mapp Cockpit
- Accessing the web-based mapp Cockpit HMI application
- Interacting with mapp components in the web-based HMI application

5 Components of the motion control system

Configuration objects are managed and accessed in the same way, regardless of the hardware being used. As a result, it makes little difference what type of motion control technology is in use.

The image below illustrates where individual configuration modules are stored during runtime, as well as where changes made to parameter values will have an effect.

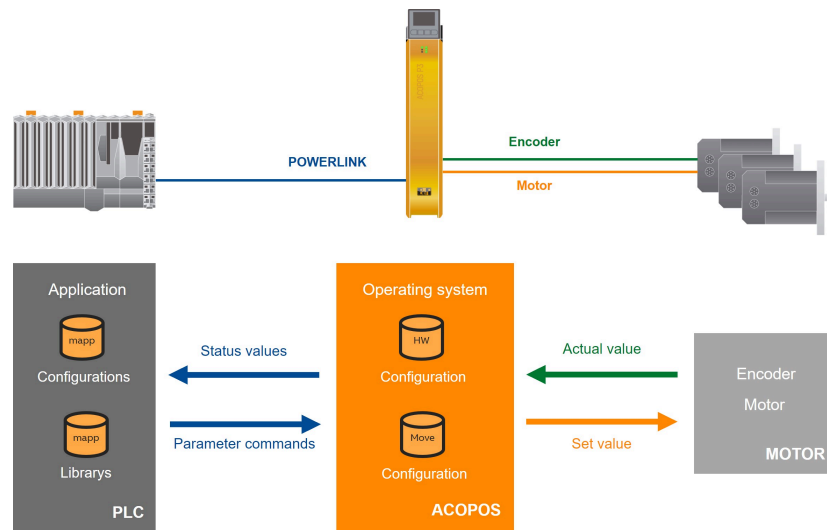


Figure 15: Overview of drive configuration components

The part of the PLC with the application (in the image) stretches over not just the programs but also the configuration objects in the Configuration View as well as the libraries used in the Logical View.

General device parameters are transferred to the ACOPOS product family devices as hardware configuration parameters (e.g. via the shortcut menu in System Designer). In addition, the movement configuration parameters from the motion function blocks for single-axis control, for example, are transferred at runtime.



Motion control \ mapp Motion \ Concept \ mapp Motion \

- mapp Axis
- mapp Motion components

Motion control \ mapp Motion \ Configuration \ Basic components \ Axis

5.1 Configuration modules

The modules added for the drive will be described in more detail in the following sections.

5.1.1 Axis configuration

The configuration file contains basic parameters for operation and defining the axis reference. The reference is later used by the software to access the axis component, e.g. for motion control and diagnostics.

The configuration file is accessed from the Configuration View. Double-clicking on the corresponding file with extension ".axis" opens a window where the initialization parameters (e.g. speed/acceleration values) of the axis components can be changed.

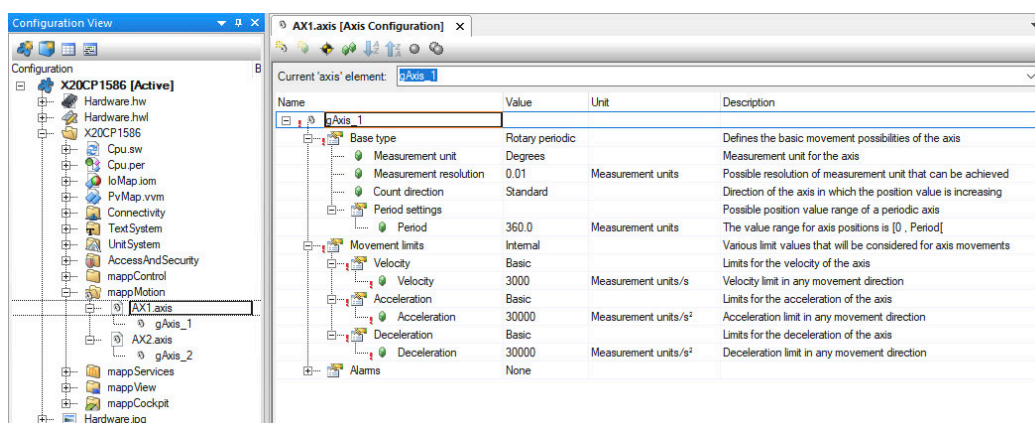


Figure 16: Configuration file of an axis component

The parameters of the axis component are arranged in clearly structured groups. These groups, the configuration options and the associated elements are described in greater detail in Automation Help.



Motion control \ mapp Motion \ Configuration \ Basic components \ Axis

5.1.2 Hardware-specific configuration

The hardware-specific configuration contains parameters that are required for the respective device used. Double-clicking on the device takes you to its configuration, as is the case with other B&R devices (e.g. X20 module). A few examples are listed below, for a complete list see Automation Help or the Automation Studio project.

- Axis reference of the axis to be operated (real axis, motor control)
- Controller settings
- Ratio of the gearbox
- Stop reactions
- Digital inputs
- Axis features

80VD100PD.C000-01 [Configuration]		
Name	Value	Description
80VD100PD.C000-01		
POWERLINK configuration		
Drive configuration		
Module		Contains parameters which influence or are relevant just for the whole module
Encoder		
Channel 1		
Real axis		
Axis reference	gAxis_1	Name of the referenced axis component
Mechanical elements		Parameter of hardware elements situated between motor encoder and load which influence the scaling
Gearbox		Specifies a gearbox by defining the ratio between a gearbox input and output
Rotary to linear transformation		Specifies a transformation factor between the output of the gear and the actual load movement
Encoder link	One encoder	
Controller		Axis controller parameters
Mode	Position controller	Mode of the axis controller
Position		Position controller parameters
Speed		Speed controller parameters
Loop filters		Parameters of the loop filters
Homing		Homing mode and parameters which can be used within the application program as preconfigured setting
Stop reaction		Reactions of the axis in case of certain stop conditions
Movement error limits		Limit values that result in a stop reaction when exceeded
Jerk filter	Used	Jerk filter
Digital inputs		Various digital input functionalities e.g. like homing switch or triggers
Simulation		Parameters which influence the simulation possibilities of this axis
Axis features		Features for an axis
Virtual axis	Not used	
Channel features		Features for the channel of a module
Channel 2		

Figure 17: Hardware-specific configuration of an ACOPOSmicro



Motion control \ mapp Motion \ Configuration \ Hardware \

5.1.3 Overall configuration

The hardware-specific settings and the axis configuration together are a functioning unit. The parameters for operation are divided into 2 clearly structured groups: Software object axis and hardware. Together they enable motor movement.

An overview in Automation Help shows which configuration options are available. An axis only needs an axis configuration to be put into operation; this configuration must be assigned to a hardware device later on.

Axes can also be combined to form an axis group so they can be operated together.

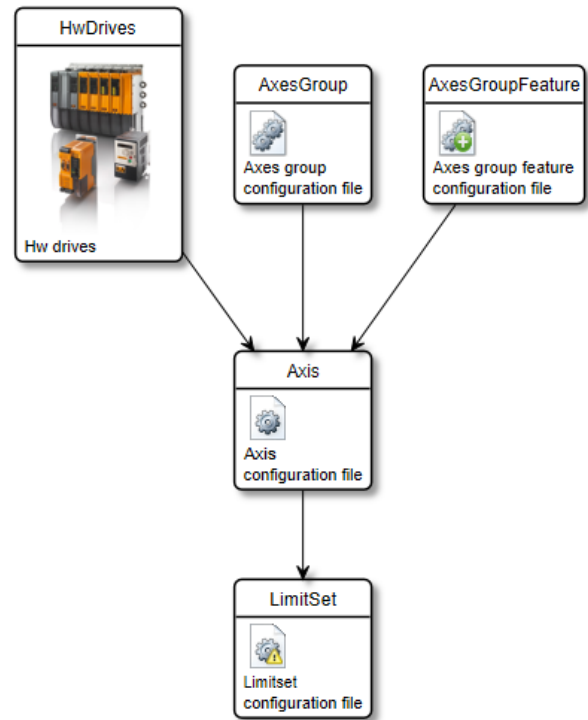


Figure 18: Overview of the mapp Motion axis configuration files



Motion control \ mapp Motion \ Configuration \ Basic components

Motion control \ mapp Motion \ Concept \ mapp Motion \ mapp Motion components \ Axis group

5.2 Configuration management

The required parameters are configured at the appropriate places in Automation Studio.

Axis component-relevant parameters are managed in the Configuration View.

Hardware-specific parameters are managed in the hardware configuration of the respective device.

When initializing mapp components after controller startup, all necessary parameters are written to the drive.

5.2.1 Axis initialization and configuration

Initialization parameters

A mapp Axis configuration file already contains default values when added. These default values, which configure the axis component, can be adjusted individually. All changes to the configuration are loaded onto the drive after startup of the controller.



Example rotary table application:

A pivoting carrier must move a product to different stations for processing (specific angular positions within a full rotation of 360°).

Positioning accuracy must be within 0.1°. MpAxisBasic is used together with command "MoveAbsolute" to approach the positions.

To make this procedure a little easier, the position is specified in degrees (with one decimal place):

```
BasicParameters.Distance := 135.0; // Target position 135°
```

The carrier is driven by a gear (gear ratio= 5:1) using a servo motor.

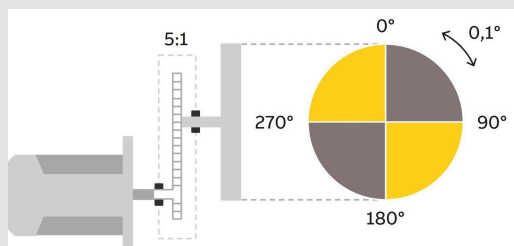


Figure 19: Sketch of the mechanical structure



In the configuration of the axis and the device-specific configuration, the unit of measurement, measurement resolution, gear ratio and axis period must be specified.

The following configuration serves as a solution for the example:

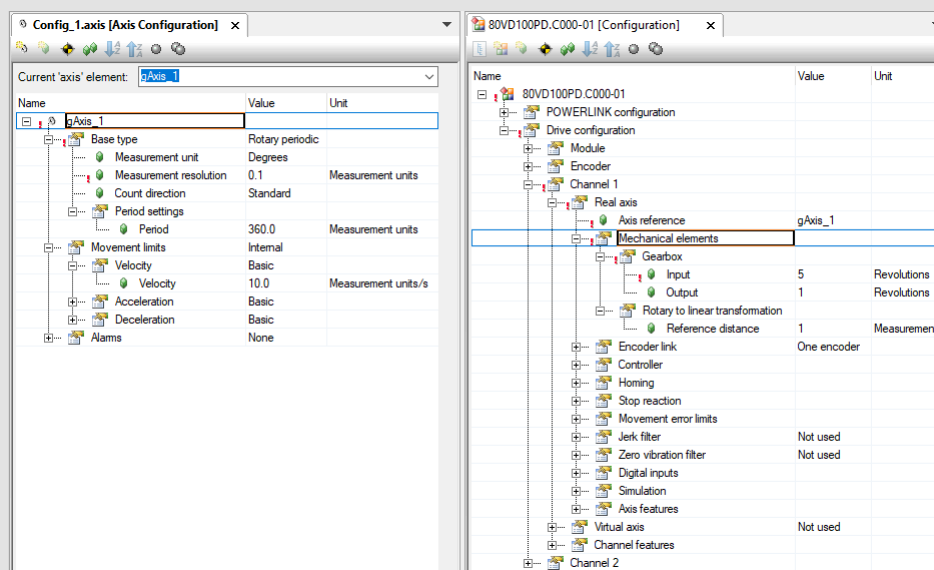


Figure 20: mapp Motion configuration for the rotary table example (axis and device-specific configuration)



Commissioning this configuration results in the following correlations:

Degrees [°] are defined as the unit of measurement. The accuracy of the resolution is 0.1°. One period (rotation) equals 360°. The gear ratio was specified as 5:1. A rotation on the gear output equals a rotation on the rotary table.

The reference distance remains unchanged because it is not taken into account when "Degrees", "Gradians" or "Revolutions" are used as the measurement unit.

In this case, the following value is used internally:

Degrees: 360

Gradians: 400

Revolutions: 1

Exercise: Configure the unit system for a spindle drive

A gripper is positioned using a spindle drive connected to a motor via a gearbox. The gear ratio is 5:1 ($i=5$). For each motor rotation on the gearbox output, there is a feed rate of 0.6 mm. The positioning should be exact to 0.1 mm. The position setpoint is entered in mm*.

Determine the settings for unit of measurement, measurement precision, gear ratio and reference distance in the mapp Motion configuration.

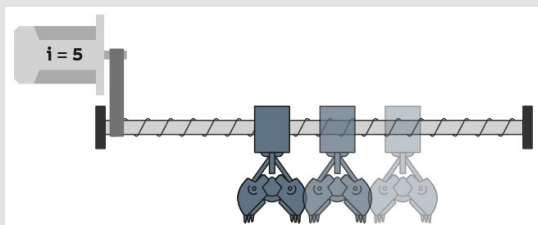


Figure 21: Schematic illustration of a spindle drive

- 1) Determine and configure suitable units of measurement.
- 2) Configure the measurement precision.
- 3) Configure the gear ratio.
- 4) Configure the reference distance.
- 5) Configure the limit values and pay attention to the maximum speed of the motor, for example.
- 6) Configure the software limits in a range from -5 mm to 205 mm.
- 7) Add MpAxisBasic.
- 8) Save and transfer the changes.
- 9) Execute movements and check results.



It should be taken into consideration that the parameters for maximum speed, lag error, acceleration and software limits should also be changed in the appropriate ratio when changing the axis parameter units.



Entry – Unit for traversing distance	Number of motor revolutions
0.6 mm	5 revolutions
2.4 mm	20 revolutions
90.0 mm	750 revolutions

Table 4: Correlation between the input value of the traversing distance and the number of motor revolutions

* Entries such as 0.1 mm or 240 mm traversing distance are permitted.



Settings:

- Base type = Linear bounded
- Measurement unit = Millimeters
- Measurement resolution = 0.1
- Lower limit = -5
- Upper limit = 205
- Gearbox input = 5
- Gearbox output = 1
- Reference distance = 0.6

Config_1[axis Configuration] x			
Current 'axis' element: gAxis_1			
Name	Value	Unit	Descr
gAxis_1			
Base type	Linear bounded		Definer
Measurement unit	Millimeters		Measur
Measurement resolution	0.1	Measurement units	Posibl
Count direction	Standard		Directi
Movement limits	Internal		Varout
Position			Mover
Lower limit	-1000.0	Measurement units	Lower
Upper limit	1000.0	Measurement units	Upper
Velocity			Limits fi
Basic			Velock
Velocity	5	Measurement units/s	Limits fi
Acceleration			Accele
Basic			Limits fi
Acceleration	50.0	Measurement units/s²	Decele
Deceleration	50.0	Measurement units/s²	

ACOPOS_any [Configuration] x			
Name	Value	Unit	
ACOPOS_any			
POWERLINK configuration			
Drive configuration			
Module			
Encoder			
Channel 1			
License	Single		
Real axis			
Axis reference			
Mechanical elements			
Gearbox			
Input	5	Revolutions	
Output	1	Revolutions	
Rotary to linear transformation			
Reference distance	0.6	Measurement units/Gearbox output revolution	
Encoder link	One encoder		
Controller			
Homing			
Stop reaction			
Movement error limits			
Jerk filter	Not used		
Digital inputs			
Simulation			
Axis features			
Virtual axis	Not used		
Channel features			
Channel 2			
Channel 3			

Figure 22: mapp Motion configuration for the spindle drive (axis and device-specific configuration)

The following correlations result from the settings:

6 Commissioning and diagnostics

Automation Studio and web-based applications both offer several tools for diagnostics.

- SDM – System Diagnostics Manager
 - mapp Cockpit
 - B&R Scene Viewer
- Standalone program that can be downloaded from the B&R website, e.g. to view complex movements (such as those of a robot system)
- [6.3.1 "Logger" on page 30](#)
 - [6.3.2 "Network command trace" on page 30](#)
 - [6.3.3 "Trace" on page 31](#)



Figure 23: Commissioning and diagnostics

System Diagnostics Manager

System Diagnostics Manager is accessed via a web browser (preferably Google Chrome) and via the IP address of the controller. The integrated web server must be enabled in Automation Runtime. In SDM, for example, the Logger can be read out and the status of the hardware checked.

Possibilities of SDM:

- Read-only access to the controller
- Download network command trace.
- Download the Logger.
- Complete diagnostic image with System Dump.



Diagnostics and service \ Diagnostics tools \ System Diagnostics Manager

mapp Cockpit



mapp Cockpit is delivered as a Technology Package upgrade. The version of mapp Cockpit must match the mapp Motion version. The version can be downloaded from the B&R website or via dialog box "Upgrades" in Automation Studio.

A connection between Automation Studio, controller and drive is required in order to use mapp Cockpit.

mapp Cockpit offers a wide range of functions that are very helpful when commissioning or troubleshooting a drive axis or axis group.

Functions of mapp Cockpit:

- Preparing a drive and issuing commands
- Accessing and managing drive parameters
- Viewing drive status values
- Recording drive parameters and process variables
- Viewing the command sequence to and status information from the drive



Figure 24: mapp Cockpit start page.

	Start page The start page is the entrance area of mapp Cockpit. The version used is shown in the lower left corner with the mapp Cockpit in the background.
	Components Area "Components" provides an overview of the components available on the target system. It is possible to access the various available views of the components.
	Trace The trace area provides options for configuring and analyzing traces. Existing traces are listed in the trace overview.
	Tools The Tools area provides access to additional functions that cannot be directly assigned to any component.
	Login and logout The Login/Logout area is accessible from any point in the web-based mapp Cockpit HMI application via its icon. In this area, you can log in and out with the users configured in the user role system. To be granted write access, your user must be assigned the role with write permission.
	Diagnostics and service \ mapp Cockpit

6.1 Commands

An axis is controlled via commands.

Commands for different applications:

- Autotuning
- Preparation
- Movements and coupling
- Read and write parameters

Autotuning

The controller parameters are determined via integrated autotuning. This means that the ACOPOS servo family device and the motor are adjusted to the load. Higher load also means higher controller values, which are necessary to be able to perform a proper movement and to prevent lag errors (difference between setpoint and actual value of the position) when starting the movement.

Preparation and movement

Before you can move an axis, you must first activate the drive's controller using command **"Power On"**. The axis is homed using command **"Home"**. Movements (single axis) or couplings (one axis depending on others) can then be performed.

Read and write parameters

Command "Process ParID" can be used to read individual parameters from the drive or write them to it. However, it is recommend to always change the configurations within the Configuration View.

Exercise: Read ParIDs**Determine the following drive parameters:**

- Motor ambient temperature (ParID: 668; TEMP_MOTOR_AMB)
- DC bus voltage (ParID: 390; UDC_NOMINAL)
- Actual position (ParID: 111; PCTRL_S_ACT)
- Current torque (ParID: 277; TORQUE_ACT)
- Lag error (ParID: 112; PCTRL_LAG_ERROR)

- 1) Search for ParIDs in Automation Help



Motion control \ ACP10/ARNCO \ Reference manual \ ACP10 \ ACOPOS parameter IDs \ Overview

- 2) Enter and confirm the ParID in "Read ParID" in mapp Cockpit



The value of the parameters is calculated directly by the drive and displayed in the message area of mapp Cockpit.

Severity	Time	Description	ID
i	2019-10-02 11:14:33.478000	ParID 112 (PCTRL_LAG_ERROR) has value 7.585857	1080297501
i	2019-10-02 11:14:33.408000	Command: ProcessParID finished	1080214353
i	2019-10-02 11:14:33.378000	Command: ProcessParID	1080213516

Figure 25: Reading lag errors in mapp Cockpit

Exercise: Work out movement types

The goal of this exercise is to move an axis and independently work out the differences.

- 1) Open mapp Cockpit
- 2) Prepare axis (switch on and perform homing)
- 3) Identify commands for movement types and perform testing
- 4) Use Automation Help to work out differences for discussion

Exercise: Change speed during movement

The goal of this exercise is to have an axis rotate at different speeds without stopping the active movement.

- 1) Open mapp Cockpit
- 2) Prepare axis (switch on and perform homing)
- 3) Starting the movement
- 4) Change speed using commands and parameters
 - Which parameters?
 - What is the override function and how should it be used?

6.2 Drive parameters and homing

6.2.1 Drive parameters

The drive parameters are grouped by individual drive functions. If parameters were changed, then they can be applied and stored in the drive configuration.

See Automation Help for detailed descriptions of the grouped parameters and elements of the data structure.

The drive parameters can be managed both in mapp Cockpit and Automation Studio.

mapp Cockpit	Configuration area
Automation Studio	- Hardware-specific drive configuration - Configuration file (.axis)

Table 5: Management of drive parameters



Diagnostics and service \ mapp Cockpit \ Web-based HMI application \ Components \ Common view
 Motion control \ mapp Motion \ Diagnostics \ mapp Cockpit for mapp Motion components
 Motion control \ mapp Motion \ Configuration \ Hardware

Exercise: Configure the axis

The goal of this exercise is to define suitable values for operation.

24 V are available in most training situations, which results in performance restrictions. The limits are higher in a real environment. It is important to adjust the preset default values to the application.

The limit values should be defined in such a way that the lag error is not aborted when the movement starts.

- Open testing environment
- Increase limit values and execute controlled movement
- Optional: Check using a speed-torque characteristic curve based on the applied voltage of the inverter and motor components.
- Save values in the project and transfer them again

6.2.2 Homing

The homing mode for the motor can be configured based on the encoder system (e.g. multi-turn/single-turn) or connected sensors (e.g. inductive switch for homing procedure). The corresponding parameters and different homing mode options are described in Automation Help.

The homing mode can be managed both in mapp Cockpit and in Automation Studio.

mapp Cockpit	Command area
Automation Studio	Hardware-specific drive configuration

Table 6: Management of homing modes



Motion control \ mapp Motion \ Concept \ Controller \ Homing

Homing methods

The goal of this exercise is to become familiar with the most common homing variants. The homing variants that execute a movement are used to start a movement via "Start velocity" to search for the switch. After the switch has been found, the axis moves more slowly and precisely to the configured switching edge via "Homing velocity".

Exercise: Homing with a movement

- 1) Opening mapp Cockpit
- 2) Prepare axis (switch on and perform homing)
- 3) Set required parameters for switch search (either in mapp Cockpit under "InitHome" or in Automation Studio \ ACOPOS configuration \ Drive configuration \ Channel \ Homing)
- 4) Start homing
- 5) Training situation: Manually trigger the reference switch (since the torque may not be sufficient with 24 VDC power supply)
ETA light: Wait until the axis has found the sensor
- 6) Observe behavior based on help description
- 7) Complete homing



Example – Switch gate homing:

- 1) Edit the hardware-specific drive configuration.
 - Drive configuration \ Channel 2 \ Real axis \ Homing \ Mode = Switch gate
 - Drive configuration \ Channel 2 \ Real axis \ Digital inputs \ Homing switch \ Source = Digital in X2.Trigger 1
- 2) Edit the mapp Cockpit parameters.
 - Under "Init home", edit the parameters and execute Init home.
 - Under "Home", edit the parameters and execute Home.

Exercise: Homing without a movement

- 1) Opening mapp Cockpit
- 2) Set the required parameters for the switch search
- 3) Follow the circuit diagram described in Automation Help
- 4) After completion of the homing procedure, set the homing mode
- 5) Switch off hardware
- 6) Turn the axis by hand by a quarter turn (if the motor does not have a holding brake)
- 7) Switch on hardware
- 8) Perform homing and check whether the position is set correctly.
- 9) Repeat procedure starting at point 5

6.3 Diagnostics

6.3.1 Logger

In Automation Studio, the drive configuration can be diagnosed via the Logger. The Logger is the central point of contact for information and sorts error messages and status entries chronologically.

Information from mapp Motion is entered in the "Motion" logbook. This contains information about the system, function blocks, libraries and errors that have occurred.

The Logger can be saved with an online connection to the controller. It can then be sent by email, for example, to provide further information about the current application in the event of support requests.



Diagnostics and service \ Diagnostic tools \ Logger
Motion control \ mapp Motion \ Guides \ Diagnostics

6.3.2 Network command trace

The NC trace logs commands sent to and from the drive at the network level. Since purely formal commands are recorded, the recording is practically independent of the actual protocol. The status of the ACOPOS product is also recorded.

elapsed time: 00:03:43 Trace Data can be uploaded now.							
Index	Interface	Node	NC Object	Request	Time [s]	Response	Index
404	PLK[0]	1	ncAXIS 1	McAcpDrv = PAR_LIST_WRITE	2095.051		
405	PLK[0]	1	ncAXIS 1	CMD_CONTROLLER = ncSWITCH_ON	2095.051		
	PLK[0]	1				2095.060	406
	PLK[0]	1	ncMODU...			2095.090	407
408	PLK[0]	1	ncAXIS 1	PLCopen_MC_AxisState = ncMC_STATE_STANDSTILL	2095.091		
409	PLK[0]	1	ncAXIS 1	McCommand = 4	2104.921		
410	PLK[0]	1	ncAXIS 1	McAcpDrv = PAR_LIST_WRITE	2104.930		
411	PLK[0]	1	ncAXIS 1	HOMING_V_TRIGGER = 0 Units/s	2104.931		
	PLK[0]	1				2104.940	412
413	PLK[0]	1	ncAXIS 1	HOMING_A = 0 Units/s ²	2104.935		
	PLK[0]	1				2104.944	414
415	PLK[0]	1	ncAXIS 1	HOMING_TR_S_BLOCK = 0 Rev.	2104.939		
	PLK[0]	1				2104.948	416
417	PLK[0]	1	ncAXIS 1	CMD_HOMING	2104.950		
	PLK[0]	1		HOMING_S = 0 Units HOMING_MODE = ncDIRECT HOMING_MODE_BITS = 00000000 (4) fix_dir = ncOFF (3) ref_pulse = ncOFF (2) trigg_dir = ncPOSITIVE (1) start_dir = ncPOSITIVE (0) edge_sw = ncPOSITIVE		2104.960	418
	PLK[0]	1	ncMODU...			2104.960	419
420	PLK[0]	1	ncAXIS 1	PLCopen_MC_AxisState = ncMC_STATE_HOMING	2104.961		
421	PLK[0]	1	ncAXIS 1	STATUS_BITS	2104.980		
	PLK[0]	1				2104.980	422
423	PLK[0]	1	ncAXIS 1	McAcpDrv = PAR_LIST_READ	2105.010		
424	PLK[0]	1	ncAXIS 1	HOMING_OFFSET	2105.010		
	PLK[0]	1				2105.020	425
426	PLK[0]	1	ncAXIS 1	HOMING_TR_S_REL	2105.020		
	PLK[0]	1				2105.030	427
428	PLK[0]	1	ncAXIS 1	PLCopen_MC_AxisState = ncMC_STATE_STANDSTILL	2105.030		
429	PLK[0]	1	ncAXIS 1	McCommand = 259	2111.971		
430	PLK[0]	1	ncAXIS 1	McAcpDrv = PAR_LIST_WRITE	2111.971		
431	PLK[0]	1	ncAXIS 1	V_OVERRIDE = 10000 0.01%	2111.971		

Figure 26: Cutout of a network command trace

An instantaneous recording of the network command trace can be created and loaded from the target system under "Motion \ "NW command trace" in SDM. The evaluation is performed with Automation Studio.

In mapp Cockpit, the network command trace can be exported in the Tools area. This can then be analyzed in Automation Studio.



Motion control \ ACP10/ARNCO \ NC Diagnosis \ NC Trace
Diagnostics and Service \ Diagnostics tools \ System Diagnostics Manager (SDM) \ Accessing System Diagnostics Manager \ SDM – SVG pages \ SDM page – Motion / NW Command trace

6.3.3 Trace

mapp Cockpit contains the trace and live value sections for data acquisition.

With the **Trace** function, drive parameters on the drive and variables on the PLC can be recorded and displayed simultaneously.

The **live value area** contains a selection of relevant variables of the component. The current values of these variables are displayed along with their units and trends.

Drive parameters are recorded in real time directly on the drive. The data collected from the ACOPOS servo drive is loaded into mapp Cockpit via the controller and displayed graphically. A number of different tools such as reference cursors and measurement cursors can be used for detailed analysis of this data.

The Trace function offers the following options:

- Records speed, acceleration and current values
- Checks the motor load
- Checks the thermal load on the ACOPOS
- Checking the positioning processes
- Synchronous recording of parameters from multiple axes

Configure a trace.

The Trace settings are adjusted in the Trace Configuration View. Data points to be recorded, recording duration and triggering behavior are configured.

Exporting trace data

The trace data obtained can be exported in the trace analysis view. The data is stored as ".csv" on the terminal device.



Diagnostics and service \ mapp Cockpit \ Web-based HMI application \ Trace
Motion control \ ACP10/ARNCO \ Reference manual \ ACP10 \ ACOPOS parameter IDs \ Overview

Exercise: Trace recording

The goal of this exercise is to become familiar with working with Trace. Some parameters are included and can be viewed in the mapp Cockpit Trace.

- 1) Opening mapp Cockpit
- 2) Configure trace (data points, timing and trigger)

Data points:

- Actual position
- Actual speed
- Current motor torque
- Lag error

- 3) Start movement, e.g. additive movement
- 4) Evaluate trace



The figure shows that an additive movement was carried out with 4 revolutions. The speed slowly increases to 500 deg/s because the user has set acceleration to 500 deg/s². The lag error occurs between -0.004 and 0.093.

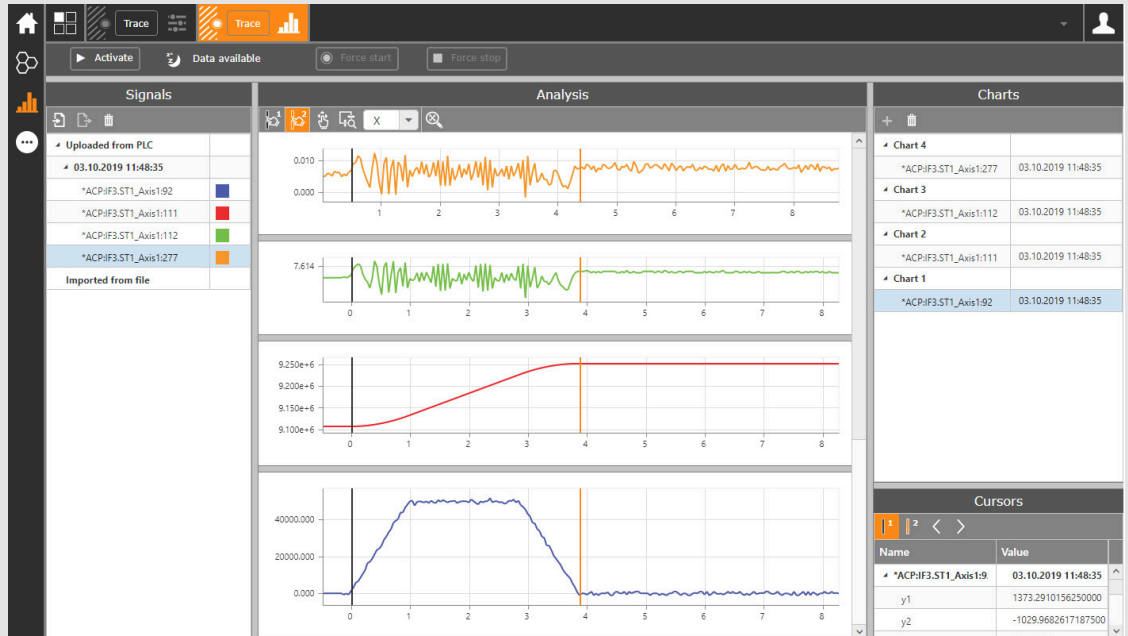


Figure 27: Orange: current torque; green: current lag error; red: current position; blue: current speed

Exercise: Record lag error using Trace function

The goal of this exercise is to abort a motion by exceeding the maximum lag error. The lag error should be recorded via the Trace function. In addition to the Trace function, other B&R diagnostic devices will also be discussed in detail. There are several ways to generate a lag error in a training situation. The axis can be blocked by holding it in place, for example. Furthermore, if the speed or acceleration is too high, this can result in a lag error⁴.

1) Prepare axis (switch on and perform homing)

2) Configure the Trace function

Data points:

- Actual speed
- Speed setpoint
- Lag error

3) Enable trace recording.

4) Perform positive movement

5) Block axis by hand

6) Check for errors:

- Trace in mapp Cockpit
- EventLog
- SDM

7) Acknowledge axis errors

3 Measurement resolution = 0.01

4 Example: The speed is increased gradually so that the maximum lag error is exceeded. Right before movement stops, the speed can be measured and saved as the maximum speed in the Automation Studio project.



The figure shows that the axis was blocked approx. 2 seconds after the movement was started. The lag error increases up to the "maximum position error" configured. The movement is aborted and the drive enters an error state.

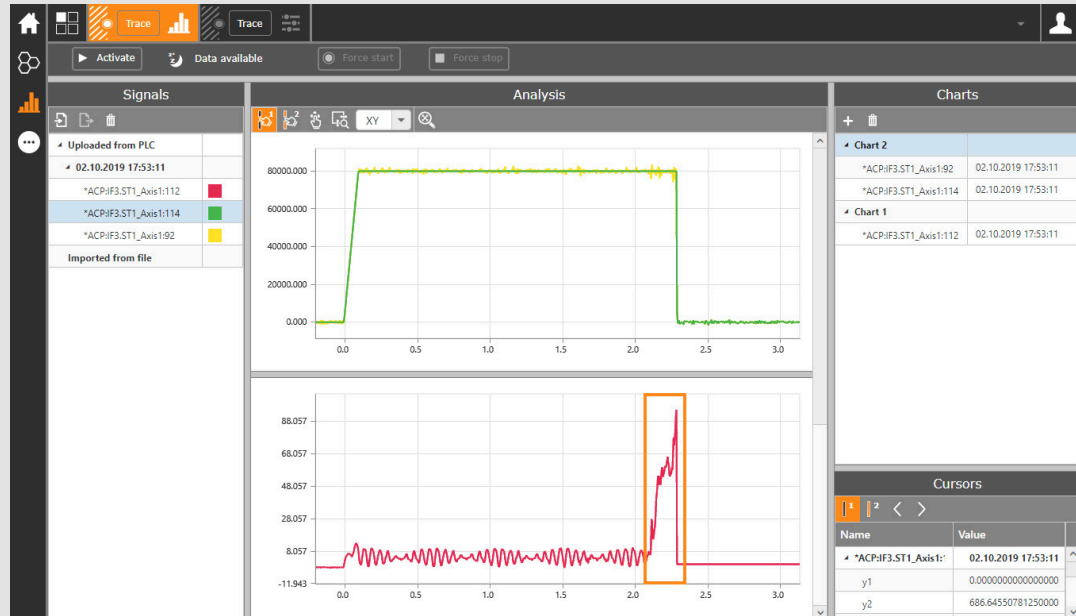


Figure 28: Red: lag error; green: speed setpoint; yellow: actual speed

Exercise: Configure the jerk time, record the movement, calculate acceleration

A jerk filter is configured in the hardware configuration to protect the drive mechanics during acceleration and braking processes.

The goal of this exercise is to compare the results with and without the jerk filter configured.

For Axis 1, a jerk filter is configured; for Axis 2, no jerk filter is configured. The same movement is executed and recorded on the two axes one after the other. The acceleration is calculated from the recorded speed. Finally, the graphs are arranged into charts and analyzed.

- 1) Configuring a jerk filter
For example, Channel 1: Jerk filter – used, Jerk time – e.g. 0.02 seconds
Channel 2: Jerk filter – not used
- 2) Switching on and homing both axes
- 3) Configure and enable the Trace function in Automation Studio or mapp Cockpit, e.g. for the data points *ACPiF3.ST1_Axis1:114 and *ACPiF3.ST1_Axis2:114.
- 4) Perform additive movement for both axes in succession.
- 5) Calculate acceleration with "Algorithm – Differentiate dy/dt ".
- 6) Arrange diagrams in mapp Cockpit.
Chart 1 with gAxis_1:PCTRL_V_SET and gAxis_2:PCTRL_V_SET, for example
Chart 2, gradient 1 and gradient 2
- 7) Analyze the recording results.



Here you can see the difference in results when a configured jerk filter is used and when no jerk filter is used.

Green = Jerk filter for Axis 1, 0.024 seconds – Speed

Red = Jerk filter "not used" for Axis 2 – Speed

Blue = Jerk filter for Axis 1, 0.024 seconds – Acceleration

Pink = Jerk filter "not used" for Axis 2 – Acceleration

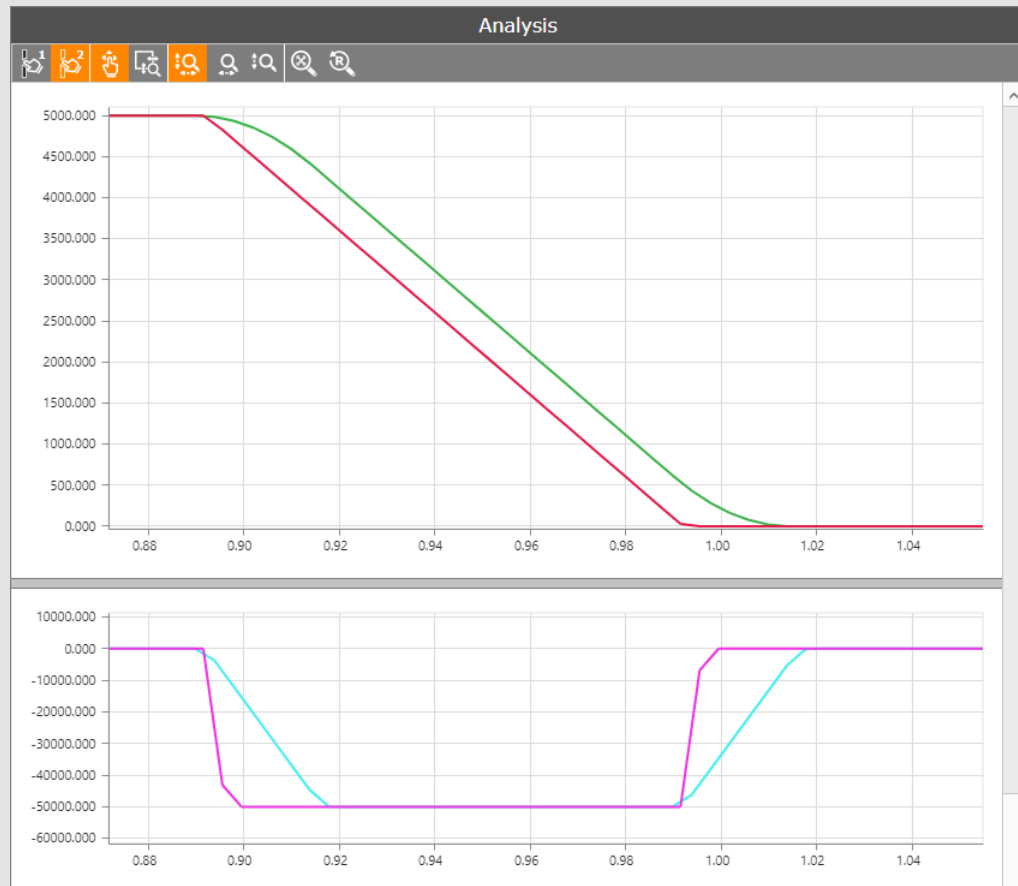


Figure 29: Comparison of the braking process of a movement with and without a configured jerk filter.

Optional exercise: Troubleshooting a communication error

The goal of this exercise is to compare it to the previous exercise that deals with lag error abort. As previously, a movement is started and the effects investigated in an error scenario.

- 1) Start a slow movement.
- 2) Disconnect the POWERLINK communication connection.
- 3) Analysis of status LEDs on the device
- 4) Reconnect POWERLINK communication
- 5) Analysis of status LEDs on the device
- 6) Stop movement
- 7) Check for errors:
 - ° mapp Cockpit
 - ° EventLog
 - ° SDM



Hardware \ Motion control \ <ACOPOS device> \ Technical data \ (<Servo drive>) \ Status indicators

6.4 Determine the controller settings using autotuning

B&R drive software is based on a cascaded control concept. A position setpoint is provided to the position controller by a setpoint generator that calculates a path profile upon receiving a positioning command. To achieve this position setpoint, the position controller specifies a speed profile. The task of the speed controller is to maintain the speed setpoint as closely as possible.

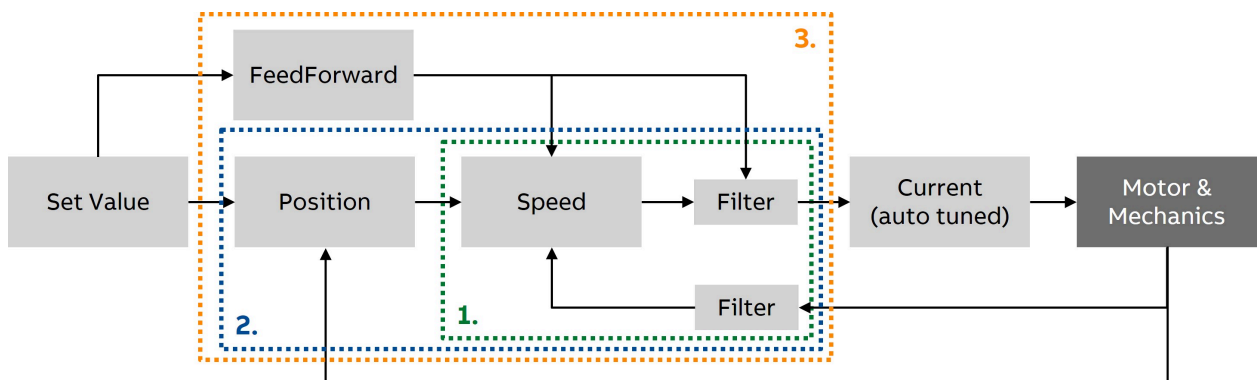


Figure 30: Simplified illustration of the cascaded control concept

The integrated autotuning procedure makes it possible to calculate the control parameters automatically. When calculating the parameters for closed-loop control, it is recommended to start with the speed controller before the position controller. The control settings must then be tested before determining the parameters for feed-forward control.

Further information about cascaded control loops:

- TM260 – The basics of closed-loop control



Motion control \ mapp Motion \ Use cases \ Autotuning
 Motion control \ mapp Motion \ Concept \ Motion control \ Closed-loop control concepts
 Motion control \ mapp Motion \ Programming \ Application program \ Libraries \ Core \ McAcpAx \
 Function blocks

Preparing autotuning

The drive must be operational before autotuning can be carried out. The functionality of the holding brake must then be checked. It is also required to check the measured direction of rotation and distance of the encoder. If any deviations are observed, or if another malfunction of the encoder is detected, the encoder must be checked both mechanically and electrically. The encoder should then be phased⁵. The tuning parameters can now be entered:

Tuning the speed controller

The speed controller's job is to determine the difference between the manipulated variable of the position controller (to which it is subordinate) and the measured speed. This calculates a manipulated variable for the subordinate current controller that works against a deviation in the speed by accelerating.

Selecting autotuning mode "Speed controller"⁶ and restarting the tuning procedure using the corresponding command will determine the parameters for the speed controller.

Tuning the position controller

The purpose of the position controller is to compare the position provided by the setpoint generator to the actual position and to generate a manipulated variable for the subordinate speed controller that works against a position change by changing the speed.

⁵ Phasing is not normally required for B&R motors. It is required, however, if the encoder has been installed at a later time.

⁶ Various filters (e.g. notch filters) are available when tuning the speed controller to stabilize the system.

Selecting autotuning mode "Position controller" and restarting the tuning procedure from the command interface will determine the parameters for the position controller.



This requires that the underlying speed controller is stable.

Testing the controller settings

Before a movement is executed with the new controller parameters, the control loop should be checked for stability. For this purpose, the system has the option of applying a short disturbance signal to the control loop ("Auto Tune Test"). If the controller parameters are correct, this disturbance disappears and the drive does not report an error.

Feed-forward components

The purpose of the feed-forward component is to reduce the load on the controller when the speed changes. The values used by the feed-forward component take the system's moment of inertia into consideration and are determined during autotuning.



This requires that the underlying speed and position controllers are stable. To do so, the axis is put into motion and must be homed.

Exercise: Determine the controller parameters using autotuning

Use the autotuning procedure to determine the controller parameters for an axis. To do so, proceed as follows:

- 1) Open mapp Cockpit
- 2) Check the holding brake and encoder signal
- 3) Perform autotuning for the speed controller
- 4) Perform autotuning for the position controller
- 5) Perform autotuning for feed-forward control
- 6) Test controller parameters
- 7) Save new controller parameters

6.5 Commissioning checklist

This section provides step-by-step instructions for commissioning a motion control system. The following points must be observed:

Safety

It is particularly important to test the safety features. This includes the emergency stop and the limit switches installed on the machine.



Figure 31: Emergency stop

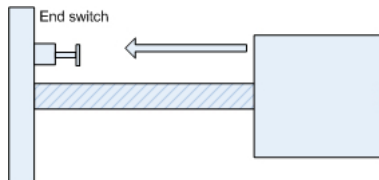


Figure 32: Limit switch



The ACOPOS user's manual and Automation Help show how to correctly wire the emergency stop and limit switch inputs.



Hardware \ Motion control \

- ACOPOS P3 \ Safety technology
- ACOPOS \ Safety technology
- ACOPOSmulti \ Safety technology
- SafeMOTION \ Safety technology
- ACOPOSmicro \ Safety technology

Digital inputs

It is important to check whether the servo drive's digital inputs have been wired according to the configured parameters. mapp Cockpit can also be used to enter states for the inputs on the ACOPOS servo drive.



Figure 33: Physical location of inputs for the ACOPOS product family

Units and movement parameters

The following configured units and settings must be checked:

- Encoder resolution and units per motor revolution
- Maximum lag error
- Software limits
- Maximum acceleration / speed values
- Stop functions
- Direction of motor rotation

Jerk filter

Using a jerk filter is recommended to prevent placing an unnecessary load on the mechanical components. This is set up using device-specific parameter "Jerk filter". It is implemented using a moving average filter over multiple position setpoints and configured in [s].

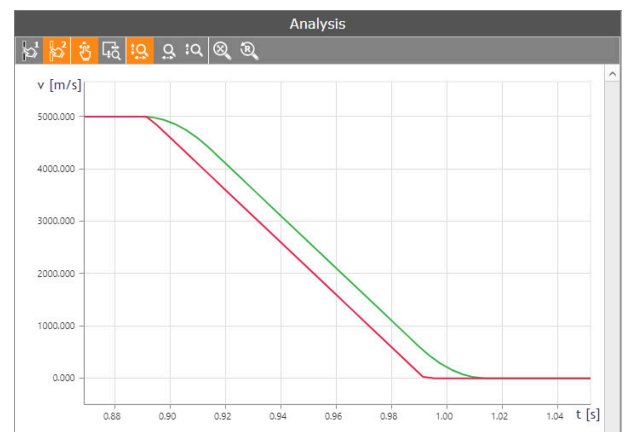


Figure 34: Effect of a configured jerk filter (green line in graph)

Homing

To ensure that positioning is accurate, a homing procedure must first be performed on the drive. There are a few different ways to do this.



Motion control \ mapp Motion \ Concept \ Controller \ Homing

Controller settings and autotuning

Controller parameters must be fine-tuned in order to adjust the control loop settings to the exact mechanical requirements. Automation Studio offers an integrated autotuning process for determining closed-loop control parameters.

Further information about control loops:

- TM260 – The basics of closed-loop control

Exercise: The effects of commissioning parameters

Change the following parameters and check their functionality.

Parameters:

- Jerk filter
- Lag error (position error)
- Direction of rotation of motor (count direction)
- Homing variant (homing mode)

- 1) Find and change the parameters in the parameter window
- 2) Save and transfer the changed values
- 3) Switch on the controller and perform a homing procedure
- 4) Start the movement and Trace
- 5) Evaluate the trace data
- 6) Perform autotuning
- 7) Start the movement and Trace
- 8) Evaluate the trace data
- 9) Work with the different homing modes

Explore the homing modes using Automation Help as a guide. Conclude by performing a homing procedure to a limit switch.

6.6 Speed-torque characteristic curve

It is helpful to take a closer look at information regarding the current requirements under which the machine is being operated.

This includes the current voltage on the ACOPOS device's DC bus since this affects the speed-torque characteristic curve of the motor. This can be viewed either on the B&R website, in Automation Help or in Automation Studio with dynamic content.

Access in Automation Studio

- This requires a motor added to a device in the ACOPOS servo family.
- Right-click on the motor and select "Speed Torque Chart".
- Enter the voltage and taking reading using the mouse pointer in the calculated diagram.

Access via the B&R website

- Open the B&R website, www.br-automation.com.
- Products → Motion control → <Type> synchronous motors → Cooling type <A, B> → Size <#> → 8LSA... (motor type)
- Select the drive with a suitable DC bus voltage and make readings using the mouse pointer in the static diagram.

Access via Automation Help

- Open Automation Help
- Examples are located in the hardware section under the motors (accessed similar to the B&R website).



Hardware \ Motion control \ <Motor> \ Technical data \ <Cooling type> \ Technical data <motor ID> \ Speed-torque characteristic curve

Exercise: Speed-torque characteristic curve of a motor

The goal of this exercise is to view a live speed-torque characteristic curve. This clearly demonstrates the limits of the 24 V used in training situations.

- A motor added to a device in the ACOPOS servo devices is important.
- Right-click on the motor and select "Speed Torque Chart".
- In Automation Studio, the notional voltage value 17 V (24 VDC in the DC bus divided by the root (2) in order to specify the notional, not-yet-rectified current value for the calculation) can be entered.
- Analyze the motor's torque curve and determine the reasons for the lag error (deviation between actual position and position setpoint) starting from a certain speed.
- For information regarding real-world applications with proscribed nominal voltage, see the B&R website or Automation Help.

7 Further drive functions

There are a large number of auxiliary functions provided by the operating system used by the ACOPOS servo devices. A few of them are listed here. For more information, see Automation Help.



Motion control \ mapp Motion \ Programming \ Application program \ Modules \ McAcpSys

2-encoder control

2-encoder control if position control should be performed on an external position encoder in an application.

reACTION Technology

Predefined function blocks can be configured remotely on the ACOPOS drive to ensure very fast reaction times or the collection of high-speed signals at very high processing speeds (400 µs).

This includes logical operations (AND, OR, etc.) as well as arithmetic calculations, comparisons, profile generators, the use of cams in switching networks and much more.

Encoder phasing

If B&R does not deliver the motor and encoder in a housing, phasing (measuring the encoder zero point in relation to the generated current phasor in the stator) is required to ensure optimal commutation.

Suppression of periodic disturbances

If impacts or resistance occurs on an axis within a motor revolution or within an interval, it is possible to compensate for these periodically occurring disturbances.

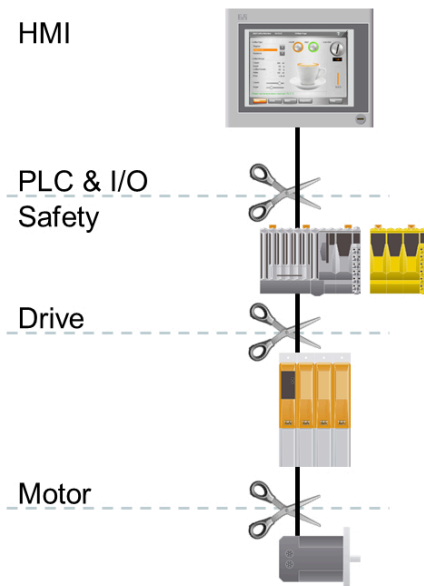
The goal is higher positioning precision and control quality with respect to the setpoint.

Controlling a B&R stepper motor axis



Motion control \ mapp Motion \ Programming \ Application program \ Libraries \ Core \ McStpAx

8 Simulation options



Automation Studio provides extensive simulation options for the controller, HMI application, drive controller, motors and even loads on the motor. In essence, all components of an integrated automation solution from B&R can be simulated.

If it is not possible or desirable to operate the actual motor on the machine, it can be simulated instead. Movement profiles can be carried out on the controller or PC, even if the entire drive system is not available. The load within defined parameters can also be simulated.

The platform-independent Automation Runtime system allows control programs to be created and tested directly on the PC. This function is also available for the safety application. Control applications can be executed in slow motion or time lapse in order to hone in on different phases of the machine's lifecycle.

Integrated VNC and web server functionality makes it possible to operate HMI applications not just remotely, but also directly on the PC.

The integrated WinIO interface makes it possible to fully simulate I/O points.

Figure 35: Complete simulation at every level

8.1 Simulation of controller and drive

Controller simulation

Simulation of a controller can be started by selecting the simulation icon in Automation Studio. All control programs run directly on the PC. This means that all of the software functions in the control application can be configured and tested independently of the hardware. When you switch to simulation mode, the project is rebuilt, the simulation environment is automatically started and an online connection to Automation Runtime Simulation is established.

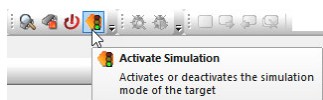


Figure 36: Activating CPU simulation from the Automation Studio toolbar

The active CPU simulation is displayed in the Automation Studio status bar with the same icon as during activation.

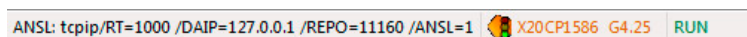


Figure 37: Automation Studio status bar – Simulation running



Drive simulation

If the drive is simulated on the actual controller CPU, this enables the drive simulation. The simulation mode can be selected in the drive's hardware configuration.

Depending on the mode, either a complete simulation is performed or only setpoint generation without controller cascading. After changing the simulation mode, the project must be built and sent to the controller.

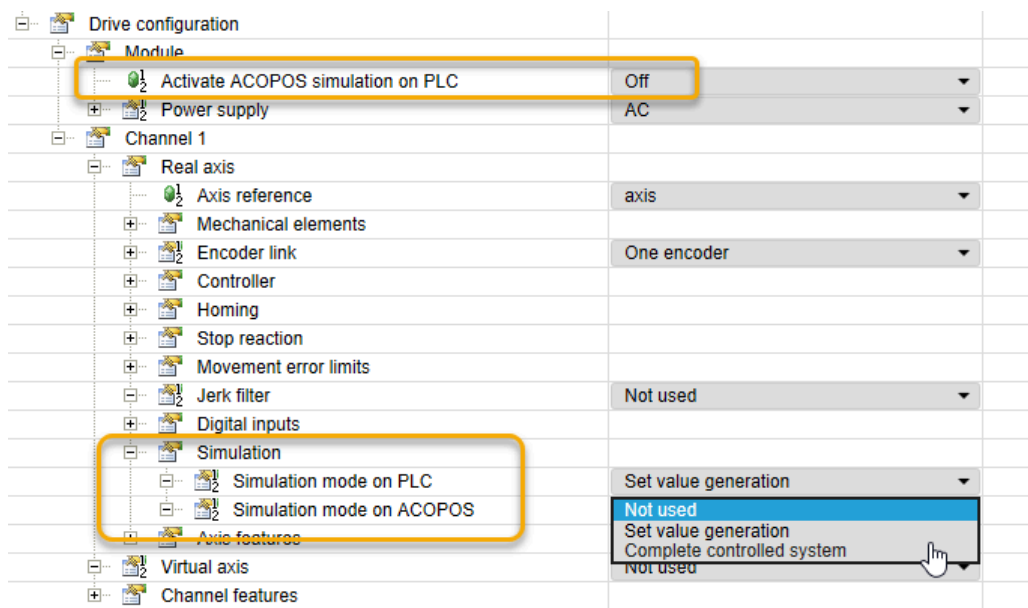


Figure 38: Enabling simulation in the device's hardware configuration



Motion control \ mapp Motion \ Configuration \ Hardware
Motion control \ mapp Motion \ Concept \ Controller \ Simulation

Simulation of the load or motor

To simulate the motor when only the PLC and ACOPOS drive are available, the appropriate settings must be made in the device-specific configuration.

Various settings are possible. If a load is not specified, only the motor is simulated.

The load can also be simulated and enabled on devices in the ACOPOS servo family.

Drive configuration			
Module			
Activate ACOPOS simulation on PLC	Off		Contains parameters which influence or are relevant just for the whole module Activates or deactivates the ACOPOS simulation on the PLC
Power supply	AC		
Encoder			
Encoder X41			
Encoder X42			
Encoder X43			
Channel 1			
License	Single		Defines whether to use single function licenses or a flat license for this channel
Real axis			
Axis reference			Name of the referenced axis component
Mechanical elements			Parameter of hardware elements situated between motor encoder and load whi
Encoder link	One encoder		
Motor and position encoder	Encoder X41		
Controller			Axis controller parameters
Homing			Homing mode and parameters which can be used within the application progra
Stop reaction			Reactions of the axis in case of certain stop conditions
Movement error limits			Limit values that result in a stop reaction when exceeded
Jerk filter	Used		Jerk filter
Jerk time	0.002	s	Jerk filter time
Digital inputs			Various digital input functionalities e.g. like homing switch or triggers
Simulation			Parameters which influence the simulation possibilities of this axis
Simulation mode on ACOPOS	Complete controlled system		Parameters for the motor and load simulation on the drive
Load model	One mass based on motor parameters		Parameters of the load simulation model
Additive load PartID	One mass based on motor parameters		1-mass load model automatically calculated based on motor parameters
Axis features	One mass		Features for an axis
Feature reference 1	Two masses		Name of the axis feature reference
Virtual axis	Not used		
Channel features			Features for the channel of a module
Feature reference 1			Name of the axis feature reference

Figure 39: Load simulation configuration in the device-specific settings

Exercise: Simulation

The goal of this exercise is to work with the different simulation options.

As a result, the different possibilities should be tried out in this exercise.

- Switch motor simulation on/off in the hardware configuration
- Switch on/off device simulation in the hardware configuration
- Locate the settings for configuring the load simulation (device-specific parameter)
- Simulate the controller and the drives in Automation Studio using "Online / Enable simulation".

9 Summary

Drives are added in Automation Studio in the Physical View or in System Designer. The drive configuration can be added easily to the Configuration View from the Toolbox.

The axis component can be put into operation quickly using mapp Cockpit.

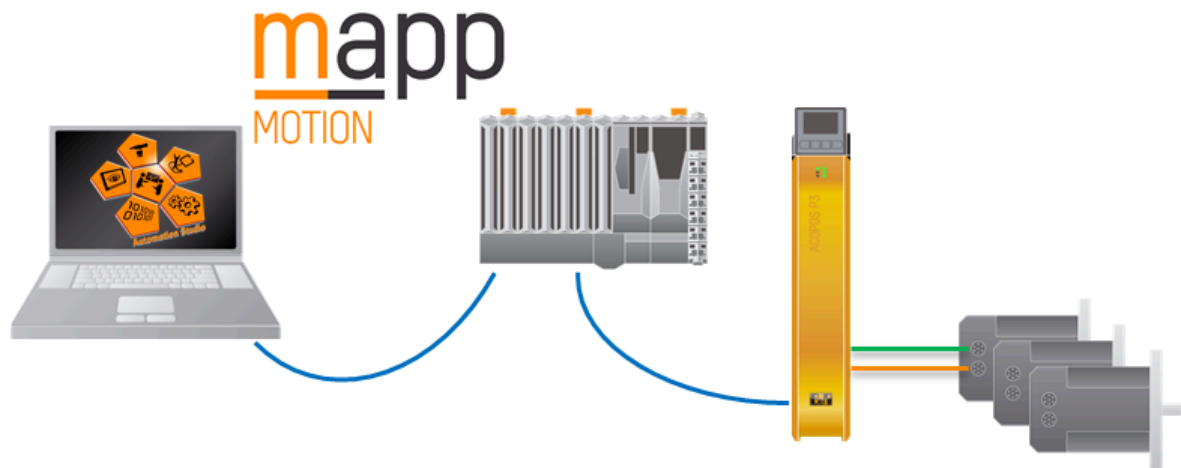


Figure 40: The integrated drive concept in Automation Studio

The Logger makes it possible to monitor drive communication and provides information about called function blocks. The integrated autotuning function allows controller settings to be determined very quickly. System Diagnostics Manager is used to read basic information about the control system even without Automation Studio. Automation Help provides extensive support for drive configuration, diagnostics and installing drive components. The powerful simulation environment provided by Automation Studio allows the implementation and testing of drive applications right at the workstation, further helping to reduce the amount of time needed for commissioning.

Automation Academy

Gain additional knowledge

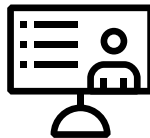
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B&R
Industrial Automation GmbH
A member of the ABB Group
B&R Straße 1
5142 Eggelsberg, Austria
office@br-automation.com

t +43 7748 6586-0
f +43 7748 6586-26

br-automation.com

